

Data-Driven Stock Analysis using Time Series Models on StockGro

Time Series Analysis 2026 Capstone Project

Submitted By

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1. Introduction

This capstone project focuses on applying time series forecasting techniques for stock market analysis and portfolio construction using the StockGro virtual trading platform. The project integrates quantitative analysis, forecasting models, volatility estimation, and portfolio allocation strategies to simulate a real-world investment workflow.

The objective of the project was to:

- Analyze historical NSE stock data
- Build forecasting models using ARIMA, Prophet, and LSTM
- Estimate volatility and trend behavior
- Construct a diversified virtual portfolio of ₹10,00,000
- Execute the strategy live on StockGro
- Compare model predictions with actual market outcomes

The project combines data science, financial forecasting, and portfolio management concepts in a practical market simulation environment.

2. Stock Universe Selection

To ensure diversification across multiple sectors, five NSE-listed stocks were selected from different industries.

Stock	Sector	Reason for Selection
Reliance Industries	Energy & Conglomerate	Large-cap stock with strong market influence and stable historical trend

Stock	Sector	Reason for Selection
Infosys	Information Technology	High liquidity and strong sector momentum in IT
HDFC Bank	Banking	Stable banking stock with moderate volatility
Tata Steel	Metals	Volatile stock useful for trend and volatility analysis
Sun Pharma	Pharmaceuticals	Defensive sector stock with consistent market presence

Selection Rationale

The stock selection was based on:

- Sector diversification
- Rolling volatility analysis
- Trend consistency from historical data
- Correlation balancing across sectors

This diversified approach helped reduce concentration risk and improved portfolio stability.

3. Dataset and Data Collection

Data Source

Historical stock market data was collected using the Yahoo Finance API through the `yfinance` Python library.

Dataset Specifications

Parameter	Value
Source	Yahoo Finance
Time Period	Jan 2021 – Dec 2025
Interval	Daily (1 Day)
Market	NSE

Libraries Used

- pandas
- numpy
- matplotlib
- seaborn

- yfinance
 - prophet
 - tensorflow
 - scikit-learn
 - statsmodels
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4. Data Preprocessing

Several preprocessing techniques were applied before building forecasting models.

Preprocessing Steps

4.1 Handling Missing Values

Missing values were handled using forward filling and removal of invalid rows.

4.2 Stationarity Testing

The Augmented Dickey-Fuller (ADF) test was used to check stationarity.

ADF Test Results

Stock	Result
Reliance	Non-Stationary
Infosys	Non-Stationary
HDFC Bank	Non-Stationary
Tata Steel	Non-Stationary
Sun Pharma	Non-Stationary

Differencing was applied where required to stabilize the mean and variance.

4.3 Train-Test Split

Dataset	Period
Training Set	Jan 2021 – Jun 2025
Testing Set	Jul 2025 – Dec 2025

4.4 Scaling

MinMaxScaler was used for LSTM preprocessing.

5. Forecasting Models

Three forecasting models were implemented:

1. ARIMA
2. Facebook Prophet
3. LSTM Neural Network

5.1 ARIMA Model

ARIMA was used for statistical time series forecasting. Model tuning was performed using AIC and BIC values.

Advantages

- Interpretable
- Effective for linear trends
- Good statistical foundation

Limitations

- Struggles with non-linear patterns
- Sensitive to parameter tuning

5.2 Facebook Prophet

Prophet was used for trend and seasonality forecasting.

Advantages

- Handles seasonality well
- Easy implementation
- Effective for business forecasting

Limitations

- Lower short-term prediction accuracy in volatile markets

5.3 LSTM Model

LSTM (Long Short-Term Memory) neural network was implemented for deep learning-based forecasting.

Advantages

- Captures non-linear relationships
- Learns sequential dependencies
- Performs well on large datasets

Limitations

- Computationally expensive
 - Requires scaling and tuning
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6. Model Evaluation and Comparison

The forecasting models were evaluated using:

- MAPE (Mean Absolute Percentage Error)
- RMSE (Root Mean Squared Error)
- Directional Accuracy

Model Comparison Table

Stock	ARIMA_MAPE	ARIMA_RMSE	Prophet_MAPE	Prophet_RMSE	LSTM_MAPE	LSTM_RMSE
Reliance	0.0494	80.4558	0.1086	194.4516	0.0164	30.2821
Infosys	0.0538	93.2729	0.0922	155.9325	-	-
HDFC Bank	0.0147	19.9018	0.0237	27.4764	-	-
Tata Steel	0.0517	10.9796	0.1733	31.0868	-	-
Sun Pharma	0.0328	68.6894	0.0838	153.0439	-	-

Observations

- LSTM achieved the best accuracy for Reliance.
 - ARIMA produced more stable results compared to Prophet.
 - Prophet captured trend behavior but struggled with short-term volatility.
 - ARIMA was selected as the primary forecasting model due to consistency and interpretability.
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7. Volatility and Trend Analysis

Volatility Estimation

Log returns and rolling standard deviation were used to estimate stock volatility.

Observations

- Tata Steel showed the highest volatility.
- HDFC Bank showed relatively stable movement.
- Reliance and Sun Pharma experienced moderate fluctuations.

Trend Analysis

Seasonal decomposition was applied to analyze:

- Trend
- Seasonality
- Residual noise

Findings

- Infosys showed relatively stable upward movement.
 - Tata Steel displayed strong cyclical fluctuations.
 - HDFC Bank remained comparatively stable.
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8. Portfolio Construction and Allocation

The portfolio was constructed using:

- Forecast-Guided Allocation
- Volatility-Aware Sizing
- Sector Diversification

Final Allocation Table

Stock	Allocation (₹)
Reliance	240000
Infosys	200000
HDFC Bank	280000
Tata Steel	120000

Stock	Allocation (₹)
Sun Pharma	160000

Allocation Strategy

Higher allocations were assigned to:

- relatively stable stocks,
- stocks with better forecast consistency,
- diversified sector exposure.

Volatile stocks received relatively smaller allocations.

9. StockGro Virtual Trading Execution

The portfolio was executed on the StockGro platform using the allocated virtual capital.

Final Purchased Quantities

Stock	Quantity
Reliance	173
Infosys	179
HDFC Bank	376
Tata Steel	563
Sun Pharma	86

Execution Notes

Minor allocation adjustments were made due to StockGro exposure constraints.

Screenshots of:

- portfolio execution,
- holdings,
- portfolio summary,
- and market observations

were recorded during the trading window.

10. Predicted vs Actual Market Performance

Final Portfolio Results

Metric	Value
Invested Value	₹9,98,395.83
Current Value	₹9,87,409.88
Overall Returns	-₹10,985.95 (-1.10%)

Final Stock-Wise Performance

Stock	Final Return %
Tata Steel	+1.63%
HDFC Bank	-0.16%
Infosys	-0.36%
Sun Pharma	-2.99%
Reliance	-2.52%

Observations

- Tata Steel became the best-performing stock during the observation window.
- Reliance and Sun Pharma negatively affected portfolio performance.
- Diversification reduced the impact of losses from individual stocks.
- Market volatility and short-term sentiment influenced actual movement.

11. Reflections and Learnings

This project demonstrated the practical challenges of financial forecasting and portfolio management.

Key Learnings

- Stock prices are highly volatile and influenced by multiple market factors.
- Time series models provide probabilistic guidance rather than guaranteed predictions.
- Diversification is important for reducing portfolio risk.
- ARIMA models provide stable and interpretable forecasts.
- Real market execution introduces practical constraints such as exposure limits and transaction costs.

What Could Be Improved

- Use larger datasets and more advanced deep learning models.
 - Incorporate sentiment analysis and macroeconomic indicators.
 - Use ensemble forecasting for improved prediction accuracy.
 - Develop a fully interactive dashboard for real-time monitoring.
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12. Conclusion

This capstone successfully integrated time series forecasting, volatility analysis, and portfolio construction into a real-world stock market simulation environment.

The project demonstrated the application of:

- statistical forecasting,
- machine learning,
- risk management,
- and diversified portfolio allocation.

Although the final portfolio produced a small short-term loss, the project successfully achieved its primary objective of implementing a data-driven investment workflow and evaluating forecasting performance against real market behavior.

13. References

1. Yahoo Finance API Documentation
2. Prophet Documentation
3. TensorFlow Documentation
4. Statsmodels Documentation
5. StockGro Platform
6. NSE India Market Data